

UNIT-V

□ **Natural Language Processing:**

- Syntactic Analysis
- Semantic Analysis
- Discourse and Pragmatic Processing
- Spell Checking.

□ **Learning:**

- What is learning?
- Rote Learning.
- Learning by taking advice
- Learning in Problem Solving
- Learning from Examples
- Analogy

What is NLP?

- NLP stands for Natural Language Processing, which is a part of Computer Science, Human language, and Artificial Intelligence.
- It is the technology that is used by machines to understand, analyze, manipulate, and interpret human's languages.
- It helps developers to organize knowledge for performing tasks such as translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation.

Natural Language Processing (NLP) refers to AI method of communicating with an intelligent systems using a natural language such as English.

Processing of Natural Language is required when you want an intelligent system like robot to perform as per your instructions, when you want to hear decision from a dialogue based clinical expert system, etc.

The field of NLP involves making computers to perform useful tasks with the natural languages humans use. The input and output of an NLP system can be –

- ✓ Speech
- ✓ Written Text

Components of NLP

There are two components of NLP as given

1. Natural Language Understanding (NLU)

- Understanding involves the following tasks –
- Mapping the given input in natural language into useful representations.
- Analyzing different aspects of the language.

2. Natural Language Generation (NLG)

It is the process of producing meaningful phrases and sentences in the form of natural language from some internal representation.

It involves –

- **Text planning** – It includes retrieving the relevant content from knowledge base.
- **Sentence planning** – It includes choosing required words, forming meaningful phrases, setting tone of the sentence.
- **Text Realization** – It is mapping sentence plan into sentence structure.

The NLU is harder than NLG.

Applications

Natural Language Processing (NLP) has a wide range of applications across various industries and domains. Its ability to understand, analyze, and generate human language has led to the development of numerous practical and innovative applications. Some of the key applications of NLP include:

1. Virtual Assistants and Chatbots: NLP is at the heart of virtual assistants like Apple's Siri, Amazon's Alexa, and Google Assistant. Chatbots also utilize NLP to provide automated customer support, answer inquiries, and interact with users in natural language.
2. Sentiment Analysis: NLP can be used to analyze social media posts, customer reviews, or feedback to determine the sentiment or emotional tone (positive, negative, neutral) of the content. Businesses can use this information to gauge customer satisfaction and make data-driven decisions.
3. Machine Translation: NLP powers machine translation systems like Google Translate, enabling automatic translation between different languages.
4. Text Summarization: NLP algorithms can automatically summarize lengthy documents, articles, or news stories, making it easier for users to get the main points quickly.

5. Information Extraction:NLP techniques help extract relevant information from unstructured text, such as identifying named entities, relationships, and events from news articles or research papers.
6. Named Entity Recognition (NER):NLP models can be used to identify and classify entities (names of persons, organizations, locations, etc.) in a text.
7. Question Answering Systems: NLP-based question-answering systems can understand user questions and provide relevant answers, like IBM's Watson or Chatbot-based customer support.
8. Language Translation and Localization:NLP plays a vital role in translating software, websites, and content to adapt them to different languages and cultures.
9. Voice Recognition and Speech-to-Text:NLP enables voice recognition systems like voice assistants and speech-to-text conversion, allowing users to interact with devices using spoken language.
10. Grammar and Style Checking:NLP can be used to check and correct grammar, syntax, and writing style in documents or text editors.

11. Text Classification and Categorization: NLP algorithms can automatically categorize texts into predefined topics or classes, aiding in content organization and analysis.

12. Spam Filtering: NLP helps in identifying and filtering out spam emails and messages by analyzing their content and patterns.

13. Medical NLP: NLP is used in healthcare for clinical text analysis, electronic health record (EHR) processing, medical coding, and drug discovery.

14. Financial Sentiment Analysis: NLP can be applied to financial news and social media data to analyze market sentiment and predict stock price movements.

15. Legal Document Analysis: NLP is used to process and analyze legal documents, contracts, and case files, improving legal research and document review.

Difficulties in NLU

NL has an extremely rich form and structure.

It is very ambiguous. There can be different levels of ambiguity –

- **Lexical ambiguity** – It is at very primitive level such as word-level.
- For example, treating the word “board” as noun or verb?
- **Syntax Level ambiguity** – A sentence can be parsed in different ways.
- For example, “He lifted the beetle with red cap.” – Did he use cap to lift the beetle or he lifted a beetle that had red cap?
- **Referential ambiguity** – Referring to something using pronouns. For example, Rima went to Gauri. She said, “I am tired.” – Exactly who is tired?
- One input can mean different meanings.
- Many inputs can mean the same thing.

NLP Terminology

- **Phonology** – It is study of organizing sound systematically.
- **Morphology** – It is a study of construction of words from primitive meaningful units.
- **Morpheme** – It is primitive unit of meaning in a language.
- **Syntax** – It refers to arranging words to make a sentence. It also involves determining the structural role of words in the sentence and in phrases.
- **Semantics** – It is concerned with the meaning of words and how to combine words into meaningful phrases and sentences.
- **Pragmatics** – It deals with using and understanding sentences in different situations and how the interpretation of the sentence is affected.
- **Discourse** – It deals with how the immediately preceding sentence can affect the interpretation of the next sentence.
- **World Knowledge** – It includes the general knowledge about the world.

Steps in NLP

There are general five steps –

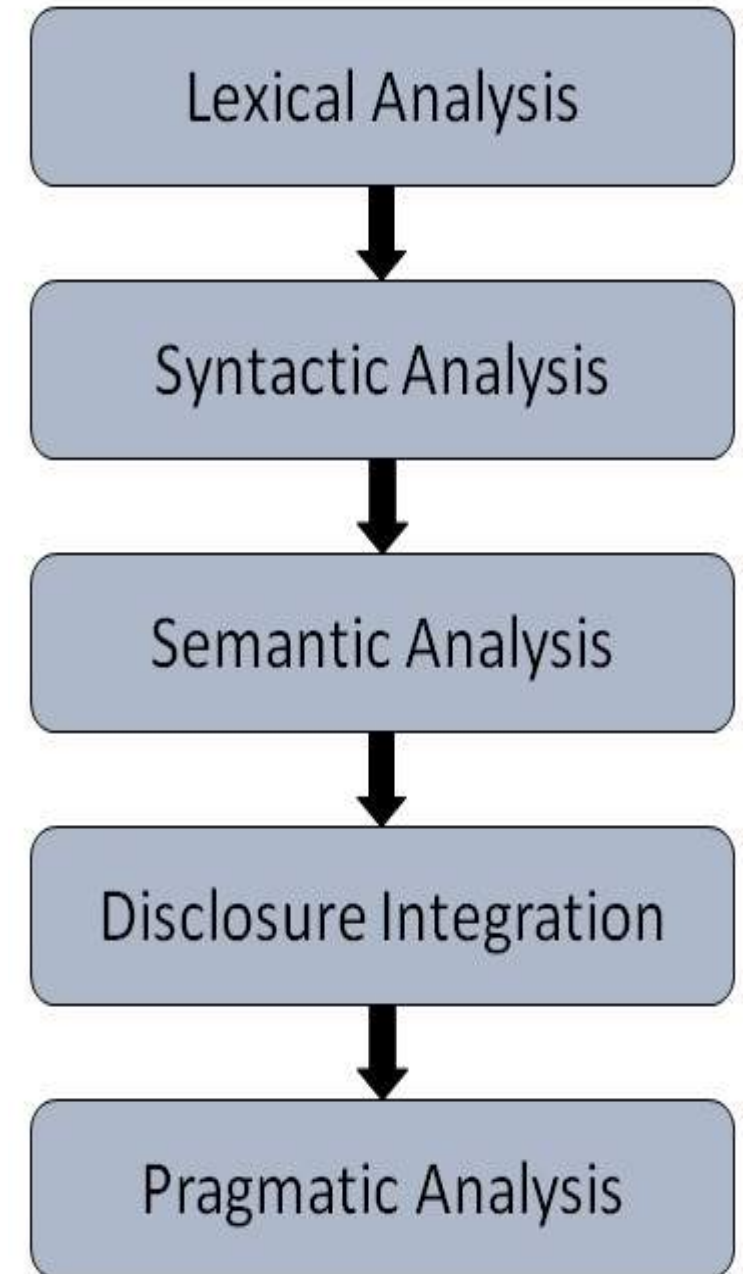
- Lexical Analysis** – It involves identifying and analyzing the structure of words. Lexicon of a language means the collection of words and phrases in a language. Lexical analysis is dividing the whole chunk of txt into paragraphs, sentences, and words.

- Syntactic Analysis (Parsing)** – It involves analysis of words in the sentence for grammar and arranging words in a manner that shows the relationship among the words. The sentence such as “The school goes to boy” is rejected by English syntactic analyzer.

- Semantic Analysis** – It draws the exact meaning or the dictionary meaning from the text. The text is checked for meaningfulness. It is done by mapping syntactic structures and objects in the task domain. The semantic analyzer disregards sentence such as “hot ice-cream”.

- Discourse Integration** – The meaning of any sentence depends upon the meaning of the sentence just before it. In addition, it also brings about the meaning of immediately succeeding sentence.

- Pragmatic Analysis** – During this, what was said is re-interpreted on what it actually meant. It involves deriving those aspects of language which require real world knowledge.



Syntactic processing

- Syntactic processing is the process of analyzing the grammatical structure of a sentence to understand its meaning.
- This involves identifying the different parts of speech in a sentence, such as nouns, verbs, adjectives, and adverbs, and how they relate to each other in order to give proper meaning to the sentence.

We have stages in syntactic analysis;

- 1) Grammar and Parsing technique
- 2) Top down vs Bottom up
- 3) Interpretations of sentences

Grammar and Parsing technique

- We need a grammar in order to parse
- Grammar=Formal specification of structure allowed in language.
- Most common way is production rule.
- Given a grammar we also need parsing technique or method of analyzing a sentence to determine its structure according to the grammar.
- The parse tree breaks down the sentence into structured parts so that the computer can easily understand and process it.
- **Context Free Grammar(CFG):**It consists of finite set of grammar rules .It is a quadruple(N,T,P,S)

N-A set of non terminal symbols.

T-set of terminal symbols.

P- set of rules left hand side production rule

S- start symbol

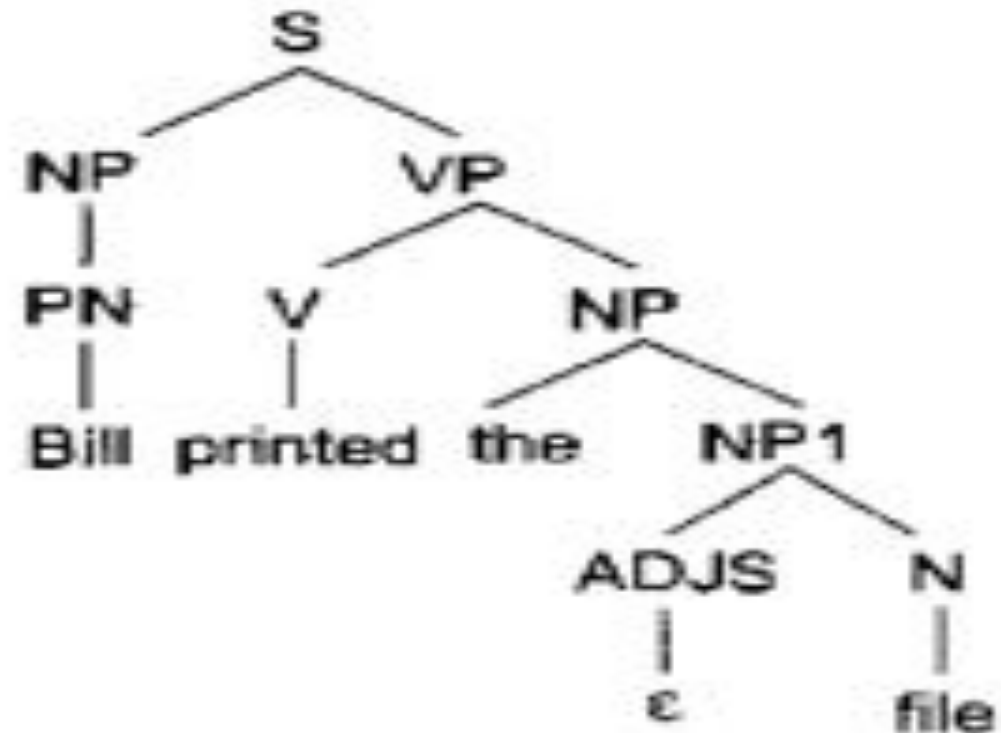
A sentence is composed of noun phase(NP) followed by verb phase(VP)

For example, consider the sentence “**The cat sat on the mat.**” Syntactic processing would involve identifying important components in the sentence such as “cat” as a **noun**, “sat” as a **verb**, “on” as a **preposition**, and “mat” as a **noun**. It would also involve understanding that “cat” is the **subject** of the sentence and “mat” is the **object**.



$S \rightarrow NP VP$
 $NP \rightarrow the NP1$
 $NP \rightarrow PRO$
 $NP \rightarrow PN$
 $NP \rightarrow NP1$
 $NP1 \rightarrow ADJS N$
 $ADJS \rightarrow \epsilon \mid ADJ ADJS$
 $VP \rightarrow V$
 $VP \rightarrow V NP$
 $N \rightarrow file \mid printer$
 $PN \rightarrow Bill$
 $PRO \rightarrow I$
 $ADJ \rightarrow short \mid long \mid fast$
 $V \rightarrow printed \mid created \mid want$

EX: Bill printed the file



Top down vs Bottom up parsing

Top down parsing:

Begin with start symbol and apply the grammar rules forward until the symbols at the terminals of the tree corresponds to the components of the sentences being parsed.

Bottom up parsing:

Begin with sentence to be parsed and apply the grammar rules backward until a single tree whose terminals are the words of the sentences and whose top node is the start symbol has been produced.

Simple top-down parse

S -> NP VP
VP -> V NP
NP -> NAME
NP -> ART N

NAME -> John
V -> ate
ART -> the
N -> cat

S
NP VP
NAME VP
John VP
John V NP
John ate NP
John ate ART N
John ate the N
John ate the cat

Simple bottom-up parse

S -> NP VP
VP -> V NP
NP -> NAME
NP -> ART N

NAME -> John
V -> ate
ART -> the
N -> cat

NAME ate the cat
NAME V the cat
NAME V ART cat
NAME V ART N
NP V ART N
NP V NP
NP VP
S

Top Down versus Bottom Up

- Similar choices in using each to forward and backward chaining in problem solving.
- Usually the branching factor is the most important consideration.
 - Is it greater going backward or forward.
- Another issue – availability of good heuristics
- Can use *bottom-up parsing with top-down filtering*
 - Essentially bottom- up parsing, but recomputed tables can be used to immediately eliminate constituents that can never be combined.

Interpretations of sentences

- Can often be more than one interpretations.
- Must decide whether to search all Possibilities or explore only a single most likely one and produce only the result of that one search.

For ex: have seen this much ,so how many paths

“Have the students who missed the exam”

Two paths:

- ”Have” is the main verb of imperative sentence
- ”Have the students who missed the exam take it today”
- Have the students who missed the exam taken it today?”

Four ways to handle this type of sentence.

- All Paths
- Best path with Backtracking
- Best path with patch up
- Wait and see

All Paths:

- Follow all possible paths and build all possible intermediate components.
- Many components will be ignored later (other input required will not appear)
- Have –if taken is not appearing.
- Inefficient because of the many spurious constituents that will be built.

Best path with backtracking:

- Follow all paths at a time, but recording decision points for later analysis if a path fails.
- Problems include a lot of time instead saving state description at each choice point even though backtracking will only occur to a few of these.
- Phrases will be reanalyzed wastefully. Choosing the wrong interpretations for have above will lead to the phrase the students who missed the exam being analyzed twice.

Best path with patch up:

- Follow one path at a time ,but when an error is detected shuffle the components that have already been formatted.
- Taking the auxiliary verb definition for “have”, the noun phrase the students who missed the exam becomes the subject of the sentence.
- If take occurs next, then interpretation is wrong so,
- The students who missed the exam was the subject of the sentence but now becomes the subject of the embedded sentence.
- The students who missed the exam take it today
- And the subject of the main sentences becomes you by default.
- More efficient approach, but requires interaction among the rules of the grammar to be made explicit in the rules for moving components.

Interpretation of sentences

- **Wait and see –**
 - follow only one path but leave making decisions about the function of each component until enough information is available to make the correct decision.
 - Efficient approach, but if the amount of lookahead necessary is greater than the buffer size, interpreter will fail.
 - Many practical systems find a single plausible interpretations (rather than all possible ones).
 - If this interpretation fails for semantic or pragmatic reasons, then a new attempt to find a different interpretation begins.

Semantic Processing

- Syntactic processing breaks the sentence into components, but does not give a meaning for the sentence.
- Main purpose is the creation of a target language representation of the sentence's meaning.
- It also imposes constraints on the representations that can be constructed and provides a way of selecting among competing syntactic analyses.
- Imposes constraints because it has access to knowledge about what makes sense in the world.

Semantic Processing

- First step is to look up the individual words in a dictionary and extract their meanings
 - Lexical Analysis.
 - Many words have several meanings, may not choose right meaning with just the word available.
 - Example:- “diamond” has the following meanings.
 - “a geometrical shape with four equal sides”
 - “a baseball field”
 - “an extremely hard and valuable gemstone”
 - Given – “Joan saw Susan’s diamond shimmering from across the room” – which meaning should be applied?

Semantic Processing

- Need to know that neither geometrical shapes or baseball fields shimmer, but gemstones do.
- Often the meaning will depend on the knowledge base being used.
- Take the word mean
 - “mean” – verb, to signify
 - “mean” – adjective, cheap or unpleasant
 - “mean” – noun, statistical average

Semantic Processing

- If we have a KB describing a statistics programs and its operation,
 - Have two distinct corresponding objects.
 - Statistical concept of a mean
 - A particular function computing the mean in this program.
 - The word needs to be mapped to one of these to understand it.
 - Lexical ambiguity is a serious problem, even in limited domains

Semantic Processing

- Lexical disambiguation – the process of determining the correct meaning
- Done by associating information about the context in which each of a word's senses may appear.
- Could mark the baseball field interpretation of diamond as a LOCATION
- Then the correct meaning in "I'll meet you at the diamond" is determined if the fact that "at" requires a TIME or a LOCATION as the object.
- Other useful semantic markers are
 - PHYSICAL OBJECT, ANIMATE OBJECT, ABSTRACT OBJECT

Semantic Processing

- Using these and the sentence “I dropped my diamond”
 - Lexical entry for “drop” specifies Physical Object as the object. Makes it easier to determine the meaning.
- However, to solve the ambiguity problem completely, more and more finely grained semantic markers need to be added.

Semantic Analysis

- Several approaches:-
 - **Semantic Grammars** combine syntactic, semantic and pragmatic knowledge into a single set of rules in the form of a grammar.
 - Result of parsing is a semantic description of the sentence.
 - Result can be used immediately with no additional processing.
 - Many ambiguities that would arise during pure syntactic parse can be avoided.
 - Syntactic issues that do not affect semantics can be ignored.
 - However, the number of rules can become very large and reduce the efficiency of the parse process.
 - Generally not accepted as a good solution
 - for large unrestricted language interfaces.

Semantic Analysis

- **Case Grammars** where the structure build by the parser also contains some semantic information.
 - Different approach where rules are written to describe syntactic rather than semantic regularities.
 - But the structures the rules produce correspond to semantic relations rather than to strictly syntactic ones.
 - Example
 - » "Susan printed the file"
 - » "The file was printed by Susan"
 - The roles of Susan and "the file" are semantically identical, but syntactically different.
 - The result using a Case Grammar would be
 - Printed(agent(Susan), object(file))

Semantic Analysis

- **Conceptual parsing** in which syntactic and semantic knowledge are combined into a single interpretation driven by the semantic knowledge.
- Syntactic parsing is subordinated to semantic interpretation.
- Another Strategy to find the structure and the meaning in one step.
 - Describes the meaning of words as conceptual dependency (CD) structures. • Similar to Case Grammars – both are driven by an expectation based on the main verb of the sentence.
 - Starts with extracting the main noun and verb.
 - More detailed than Case Grammar in that it also determines the syntactic category and the aspectual class of the verb (stative, transitive or intransitive).
 - Once the correct dictionary entry is chosen, the conceptual processor analyses the rest of the
 - sentence for components to fill the verb structure.

Semantic Analysis

- **Approximately compositional semantic interpretation** in which semantic processing is applied to the result of performing a syntactic parse.
 - Can be done either incrementally as constituents are built or all at once when a structure corresponding to a complete sentence has been built.

Discourse and Pragmatics

- To actually understand the sentence, we must consider the context in which it was uttered/ written
- To help in this, there are a number of important relationships that may hold between phrases and part of their discourse context.
- Identical entities
 - Bill had a red balloon
 - John wanted it.
- Part of entities.
 - sue opened the book she had just bought.
 - The title page was torn
- Parts of actions
 - John went on a business trip
 - He left on an early morning flight
- Entities involved in actions.
 - My house was broken onto last week. They took the TV and stereo.

Spell Checking

AI Grammar and Spell Checker refers to the use of artificial intelligence technology to automatically detect and correct grammatical errors and spelling mistakes in written text. It works by analyzing the text and comparing it to a vast database of grammar and spelling rules, as well as common errors.

Spelling errors

3 Causes

- **Insertion:** insertion of an extra letter while typing

Ex: maximum typed as mixiimum

- **Deletion:** Case of letter missing or not typed in a word

Ex: netwrk instead of network

- **Substitution: Typing** of a letter in place of the correct one as in intelilgence where in the letter I has been wrongly substituted by you.

There are 3 types in spelling error's

- **Typographic error:** these errors are those that are caused due to mistakes committed while typing
- **Orthographic errors:** A lack of comprehensions of the concerned language on part of the user.

Ex: spelling errors arithmetic ,wellcome, accomodation

- **Phonetic errors:** poor cognition on part of the listener .The word rough could be spelt as ruff and listened as lisen .

Learning

What is learning?

- According to **Herbert Simon**, learning denotes changes in a system that enable a system to do the same task more efficiently the next time.
- Arthur Samuel stated that**, "Machine learning is the subfield of computer science, that gives computers the ability to learn without being explicitly programmed".
- In 1997, **Mitchell** proposed that, " A computer program is said to learn from experience '**E**' with respect to some class of tasks '**T**' and performance measure '**P**', if its performance at tasks in '**T**', as measured by '**P**', improves with experience **E**".
- The main purpose of machine learning is to study and design the algorithms that can be used to produce the predicates from the given dataset.
- Besides these, the machine learning includes the agents percepts for acting as well as to improve their future performance.

- The following tasks must be learned by an agent.**

To predict or decide the result state for an action.

- To know the values for each state(understand which state has high or low value).
- To keep record of relevant percepts.

Why do we require machine learning?

- Machine learning plays an important role in improving and understanding the efficiency of human learning.
- Machine learning is used to discover a new things not known to many human beings.

Various forms of learnings are explained below:

1. Rote learning

Rote learning is possible on the basis of memorization.

This technique mainly focuses on memorization by avoiding the inner complexities. So, it becomes possible for the learner to recall the stored knowledge.

For example: When a learner learns a poem or song by reciting or repeating it, without knowing the actual meaning of the poem or song.

•2. Induction learning (Learning by example).

Induction learning is carried out on the basis of supervised learning.

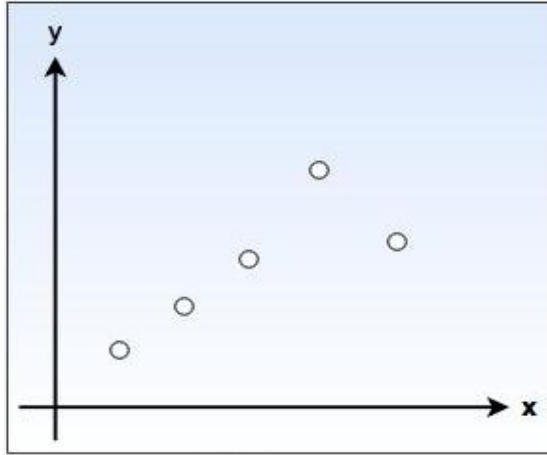
- In this learning process, a general rule is induced by the system from a set of observed instance.
- However, class definitions can be constructed with the help of a classification method.
-

For Example:

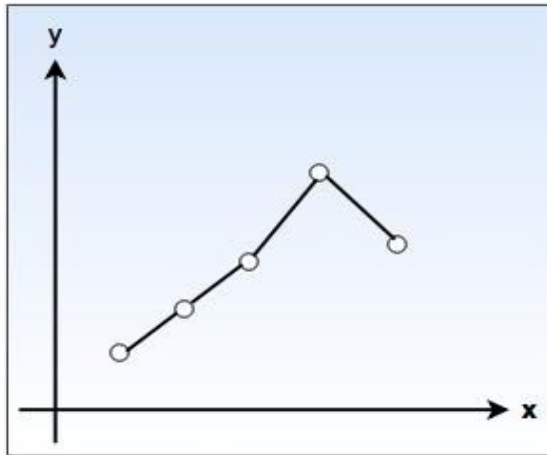
Consider that ' **f** ' is the target function and example is a pair $(x, f(x))$, where 'x' is input and $f(x)$ is the output function applied to 'x'.

Given problem: Find hypothesis h such as $h \approx f$

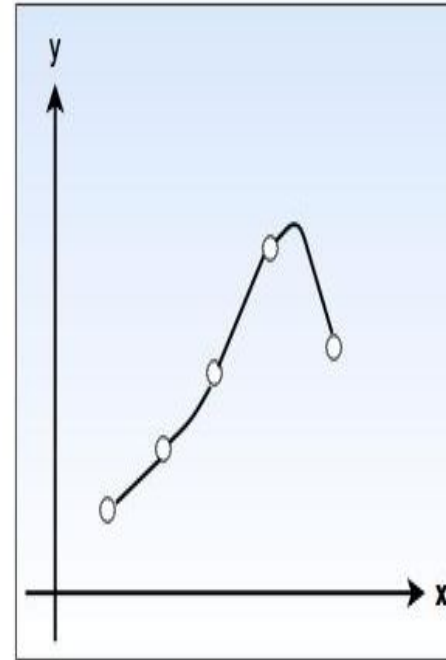
So, in the following fig-a, points (x, y) are given in plane so that $y = f(x)$, and the task is to find a function $h(x)$ that fits the point well.



Fig(a)

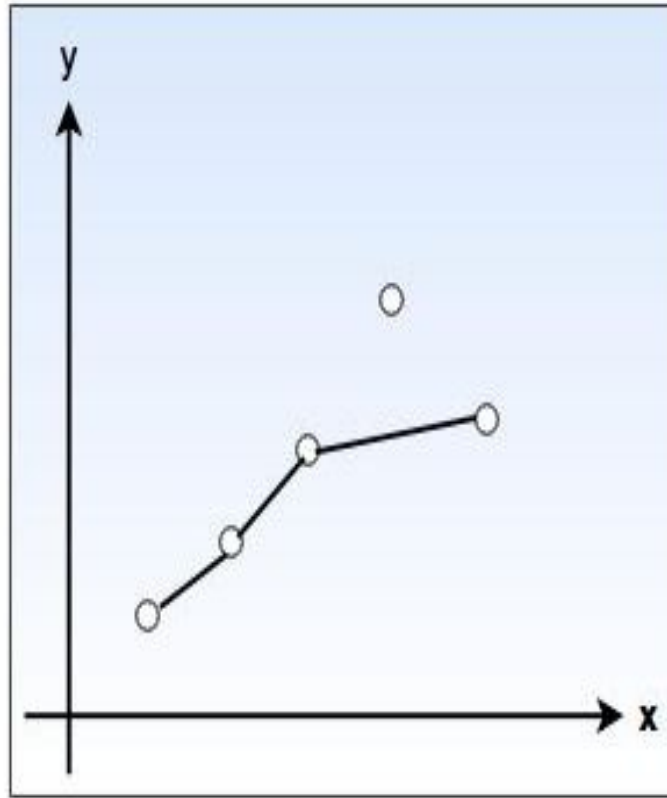


Fig(b)



Fig(c)

- Both the functions agree with the example points, but differ with the values of 'y' assigned to other x inputs.



Fig(d)

- As shown in fig.(d), we have a function that apparently ignores one of the example points, but fits others with a simple function. The true f is unknown, so there are many choices for h , but without further knowledge, we have no way to prefer (b), (c), or (d).

3. Learning by taking advice

This type is the easiest and simple way of learning.

In this type of learning, a programmer writes a program to give some instructions to perform a task to the computer. Once it is learned (i.e. programmed), the system will be able to do new things.

Also, there can be several sources for taking advice such as humans(experts), internet etc.

However, this type of learning has a more necessity of inference than rote learning. As the stored knowledge in knowledge base gets transformed into an operational form, the reliability of the knowledge source is always taken into consideration.

Explanation based learning

- Explanation-based learning (EBL) deals with an idea of single-example learning.
- This type of learning usually requires a substantial number of training instances but there are two difficulties in this:
 - i. it is difficult to have such a number of training instances
 - ii. Sometimes, it may help us to learn certain things effectively, specially when we have enough knowledge.Hence, it is clear that instance-based learning is more data-intensive, data-driven while EBL is more knowledge-intensive, knowledge-driven.
- Initially, an EBL system accepts a training example.
- On the basis of the given goal concept, an operational criteria and domain theory, it "generalizes" the training example to describe the goal concept and to satisfy the operational criteria (which are usually a set of rules that describe relationships between objects and actions in a domain).
- Thus, several applications are possible for the knowledge acquisition and engineering aspects.

Learning in Problem Solving

Humans have a tendency to learn by solving various real world problems.

The forms or representation, or the exact entity, problem solving principle is based on reinforcement learning.

Therefore, repeating certain action results in desirable outcome while the action is avoided if it results into undesirable outcomes.

As the outcomes have to be evaluated, this type of learning also involves the definition of a utility function.

This function shows how much is a particular outcome worth?

There are several research issues which include the identification of the learning rate, time and algorithm complexity, convergence, representation (frame and qualification problems), handling of uncertainty (ramification problem), adaptively and "unlearning" etc.

In reinforcement learning, the system (and thus the developer) know the desirable outcomes but does not know which actions result into desirable outcomes.

In such a problem or domain, the effects of performing the actions are usually compounded with side-effects. Thus, it becomes impossible to specify the actions to be performed in accordance to the given parameters.

Q-Learning is the most widely used reinforcement learning algorithm.

The main part of an algorithm is a simple value iteration update. For each state 'S', from the state set S, and for each action, a, from the action set 'A', it is possible to calculate an update to its expected reduction reward value, with the following expression:

where r_t is a real reward at time t , $\alpha_t(s,a)$ are the learning rates such that $0 \leq \alpha_t(s,a) \leq 1$, and γ is the discount factor such that $0 \leq \gamma < 1$.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha_t(s_t, a_t) [r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

Analogy

A comparison between one thing and another, typically for the purpose of explanation or clarification.

Example:- The analogy between the heart and a pump.

Learning by analogy

- ▶ It is a powerful inference tool.
- ▶ It generally involves abstracting details from a particular set of problems and resolving structural similarities between previously distinct problems.
- ▶ Analogical reasoning refers to this process of recognition and then applying the solution from the known problem to new problem.
- ▶ It involves developing a set of mappings between features of two instances.

Analogical Reasoning Steps

- ▶ **Retrieve:-** Retrieve cases from memory that are relevant to solving it.
- ▶ **Reuse:-** Map the solution from previous case to the target problem. This involves adapting the solution to fit new solution.
- ▶ **Revise:-** Test the new solution to real world and, if necessary, revise.
- ▶ **Retain:-** After the solution has been successfully adapted to target problem, store the resulting experience as the new case in memory.

Transformational analogy

- ▶ Suppose you are asked to prove a theorem in plane geometry.
- ▶ You might look for a previous theorem that is very similar and copy its proof, making substitutions when necessary.
- ▶ The idea is to transform a solution to a previous problem in to solution for the current problem.



Fig: Transformational analogy

Derivational analogy

- ▶ It only looks at the final solution.
- ▶ Often the twists and turns involved in solving an old problem are relevant to solving a new problem.
- ▶ The detailed history of problem solving episode is called derivation.
- ▶ Analogical reasoning that takes these histories into account is called derivational analogy.



Fig: Derivational analogy

Thank you....